Bayesian Bivariate Markov Switching Multifractal Model

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1 Project Objective

The objective of this project is to extend the Bivariate Markov Switching Multifractal Model developed by Calvet et al (2006) [1] to better capture volatility comovement for structural learning and forecasting purposes.

Markov Switching Multifractal Models (MSM) can be considered as stochastic and multiplicative variant of Fourier decomposision in a sense that it models volatility series as realizations from zero-mean Normal distribution with the variance being a product of state realizations from prespecified number of first-order two-state Markov switching processes with ordered switching probabilities (low to high). Therefore, switches in those frequencies (as in Fourier decomposition) represents regime change in short to long term expectations (such as high frequency component to short term expectations and vise versa).

For this reason, when extending MSM to model bivariate and multivariate volatilities, we need to tie switches in frequencies across series. Our model allows **switches in all frequencies across series to potentially have conditional dependence with each other** (although we expect sparsity in this structure) since we may expect change in short-term expectation in one series to correspond to that of mid/long-term for the other. For example, when considering commodity spot-futures pair, a shock in that commodity may have a lasting effect on speculation-prone futures while spot price may absorb it as a short term shock.

2 Implementations & Challenges

Analogous to the Dynamic Dependency Networks (DDN) [3], our bivariate implementation defines **one-way conditional structure** of switches where switches within one series (say β) will increase the chance of switches in frequencies within the other series α. We do this by defining the following latent logistic regression structure on switching probability of frequency i within series α:

$$logit(\pi_i) = C + B_{i,1}1_1^{\beta} + \cdots B_{i,\bar{k}^{\beta}}1_{\bar{k}^{\beta}}^{\beta},\ i = 1,...,\bar{k}^{\alpha},\ C = logit(\gamma_i^{\alpha}) - B_{i,1}\gamma_1^{\beta} - \cdots - B_{i,\bar{k}^{\beta}}\gamma_{\bar{k}^{\beta}}^{\beta}$$

where $1_{1:\bar{k}^{\beta}}^{\beta}$ represent switches of frequency 1 to $\bar{k}^{\beta}$ in series β and corresponding coefficients $B_{i,1:\bar{k}^{\beta}}$ to be the effect of switches in that specific frequency within α. The intercept C consists of base switching probability of frequcy i within α as γ_i^{α} and that of βs to ensure that when marginalizing out $1_{1:\bar{k}^{\beta}}^{\beta}$ will allow that interpretation of γ_i^{α}. Additionally, we introduced ordering of π_i's to avoid label-switching issues. As we can see, these parameters controlling the conditional dependence are not time dependent, but we could also evolve these over time. Thus, the interfrequency conditional structure is defined by multiple logistic regressions. By giving exponential priors to coefficients (we assume those to be nonnegative), we can induce sparsity on these effects as we would expect from the real data.

After coming up with this conditional structure to obtain joint distribution of frequencies, we had difficulty in obtaining the likelihood of joint states marginalizing out nuisance parameters $1_{1:\bar{k}^{\beta}}$ so that we can do Metropolis on hyperparameters.

The solution (albeit brute-force) we adopted was to compute the exact likelihood of joint states conditional on hyperparameters by considering all possible transitions of joint state $t-1$ to t with various switches with corresponding probabilies. Since state space is finite, it is computable exactly. Moreover, this exact forward step is parallelizable, so we used Spark to distribute this job to ensure efficiently in some extent (a joint model with up to 5 frequencies for both series can be run on 16G RAM Intel-i7 desktop to some mixing performance).

However, there are some numerical problems (solvable but requires some nontrivial code changes) as well as multimodality of some parameters (related to marginally tree-like structure of MSM since it is Factorial HMM) and exponential smoothing-ish parameter causing mixing issues we have trouble dealing with.

3 Simulation Results

Due to the aforementioned implementational issues, we haven't run this model on real data yet. However, we run MCMC on synthetic toy data following this bivariate MSM model to assess the severity of these various model issues in mixing results. Here we simulated series α, β with (2,3) frequencies and the switches in former is conditionally dependent on that of the latter. Thus, we have 2 latent regression with 6 coefficients in total and the result is as follows:

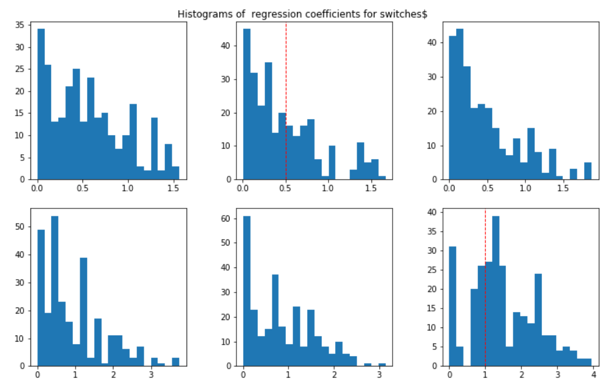

Figure 1: Posterior Regression Coefficiens

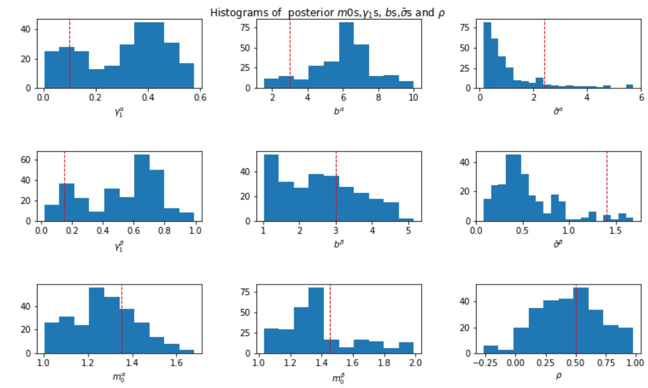

Figure 2: Posterior Hyperparameters

The figure on left is the posterior regression coefficients where histograms with red horizontal dots are those with nonzero entries. We can see that the bottom right coefficient are considered nonzero while the other one isn't. Since the top row represents the effect on lower frequency component in α, there may be not enough evidence to suggest that switches in β contributes to any significant change in switching probability in lower freqency component in α since the probability is already low. On top of that, we see from the figure on right that mixing performance on some parameters are not good (again, red dotted line represents the actual value), which probably also has an effect on the coefficients.

4 Future Work & Comments

The reason we adopted exact calculation of likelihoods is mostly due to the time constraints and I think there are sampling-driven solutions to obtain the posterior since the observations equation is normal. Other aforementioned implementational issues are also not entirely new, so should be solvable in some extent.

The obvious issue in extending this to multivariate case is the increase in number of regressors.Thus we may need some sort of scalable BMA approach such as Ma Li (2015) [2] for example.

If our future work successfully addresses all these challenges in implementing multivariate MSM, its inferential appeal is significant.**The chain of active (= sufficiently nonzero coefficients) conditional dependencies in switches may uncover some probabilistic pathway of shocks** (assuming the ordering of series which practically has some effect on the result is valid in some way) and how each series absorbs those shocks. This model of errors (fundamentally, MSM is a stochastic volatility model) combined with forecast oriented models such as DDNM or SGDLM may allow more efficient mid to long term forecasting intervals since both takes into account the graphical links of series for increased forecasting performance as well as structural learning capability.

Finally, this is a joint project with Jaeyeon Lee and my contribution includes the original idea of interfrequency modeling of Bivariate MSM, introduction of conditional dependence through latent logistic regressions and all code implementations (`https://github.com/Jantg/BEST_proj/tree/master/MSM_exact`) while Jay contributed in rationalizing the interfrequency ideas using the spot-futures examples and providing 2nd opinion on theoretical steps. We thank professor Ma and West for candid opinions and mentorship throughout this project.

5 Appendix

5.1 Full Model Specification

$$\begin{pmatrix} r_t^\alpha \\ r_t^\beta \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha & \rho\sqrt{\bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha}\sqrt{\bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta} \\ \rho\sqrt{\bar{\sigma}_\alpha^2 \prod_{i=1}^{\bar{k}^\alpha} M_{i,t}^\alpha}\sqrt{\bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta} & \bar{\sigma}_\beta^2 \prod_{i=1}^{\bar{k}^\beta} M_{i,t}^\beta \end{pmatrix}\right]$$

$$M_{i,t}^\beta = M_{i,t-1}^\beta + 1_{i,t}^\beta(\bar{m}^\beta - M_{i,t-1}^\beta),\ i = 1, ..., \bar{k}^\beta, 1_{i,t}^\beta \sim Bernoulli(\gamma_i^\beta)$$

$$M_{i,t}^\alpha = M_{i,t-1}^\alpha + 1_{i,t}^\alpha(\bar{m}^\alpha - M_{i,t-1}^\alpha), i = 1, ..., \bar{k}^\alpha,\ 1_{i,t}^\alpha \sim Bernoulli(\pi_i)$$

$$logit(\pi_i) = C + B_{i,1}1_{1,t}^\beta + \cdots B_{i,\bar{k}^\beta}1_{\bar{k}^\beta,t}^\beta,\ i = 1, ..., \bar{k}^\alpha,\ C = logit(\gamma_i^\alpha) - B_{i,1}\gamma_1^\beta - \cdots - B_{i,\bar{k}^\beta}\gamma_{\bar{k}^\beta}^\beta$$

5.2 Sampling Algorithm

1. Pick appropriate $\bar{k}^{\alpha,\beta}$ as model parameters.
2. Initialize $m_0^{\alpha,\beta}, \gamma_1^{\alpha,\beta}, b^{\alpha,\beta}, \bar{\sigma}^{\alpha,\beta}, B_{1:\bar{k}^\alpha,1:\bar{k}^\beta}$
3. Set priors on parameters: $Beta()$ for γ_1s, $Gamma()$ for bs and $Exp()$ for Bs (to induce shrinkage and we know $B_{i,j} > 0\ \forall i, j$), improper priors for others.
4. Sample Bs from normal proposal and accept it with the Metropolis acceptance rate
5. In case sampled Bs change the strict ordering of π_is ($\pi_1 < \pi_2 < \cdots < \pi_{\bar{k}}$), reject proposed Bs.
6. Sample other parameters with Metropolis reflecting random walk (to account for its domain), and compute the acceptance probability of these new set of parameters and accept/reject with that rate.
7. Back to 4 until good mixing (for better mixing performance, we can make step 4 and 6 into multiple steps as well).

References

[1] Laurent E Calvet, Adlai J Fisher, and Samuel B Thompson. Volatility comovement: a multifrequency approach. *Journal of econometrics*, 131(1-2):179–215, 2006.

[2] Li Ma. Scalable bayesian model averaging through local information propagation. *Journal of the American Statistical Association*, 110(510):795–809, 2015.

[3] Zoey Yi Zhao, Meng Xie, and Mike West. Dynamic dependence networks: Financial time series forecasting and portfolio decisions. *Applied Stochastic Models in Business and Industry*, 32(3):311–332, 2016.